

<https://doi.org/10.1038/s41746-026-02375-1>

Naturalistic facial dynamics enable quantitative clinical assessment of atypical expression phenotypes in children with autism spectrum disorder

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Existing facial-expression studies in children with autism spectrum disorder (ASD) rely mainly on discrete, task-driven measures that overlook the sustained emotional fluctuations and ambiguous expressions in naturalistic interactions. This study quantified atypical facial expression patterns in ASD during spontaneous, unscripted interactions. We analyzed 184 naturalistic video sessions from 99 children with ASD and 85 typically developing (TD) peers and extracted three features capturing spontaneous dynamics: emotion variation (temporal stability of emotional states), expression intensity (magnitude of facial muscle activation), and facial coordination (synchrony across facial muscles). These features integrated holistic and processual representations across coarse- and fine-grained levels, enabling detailed quantification of facial patterns. Compared with TD peers, the ASD group exhibited increased prominence of anger, altered emotion transition probabilities, heightened activation in non-core facial muscles, and atypical facial coordination ($p < 0.05$). These findings reveal subtle facial dynamics inaccessible to traditional approaches and provide a quantitative explanation for the hard-to-describe atypical expressions. Using the fused feature set, ASD classification reached 92.4% accuracy and 0.977 AUC. Regression analyses further predicted symptom severity with mean absolute errors of 13.94 on the ABC scale and 3.84 on the CABS scale. These quantitative and interpretable markers show promise for large-scale ASD screening in naturalistic settings.

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by social communication deficits and restricted, repetitive behaviors¹. Affecting approximately one in 100 children², ASD places substantial burdens on individuals, families, and society. Given these broad impacts, early identification is essential, as timely intervention can significantly improve long-term developmental outcomes^{3,4}.

Social impairment is a core characteristic of ASD^{5,6}, and facial expressions are a primary channel for conveying socio-emotional information, accounting for over half of emotional communication^{7–10}. Children with ASD consistently show atypical facial expressivity, including unclear or poorly regulated emotions, a tendency toward neutral or negative affect, and facial movements that may appear mechanical or contextually inappropriate^{11–13}. These behavioral patterns align with psychological

theories pointing to difficulties in emotion understanding, empathy, and emotional awareness^{14,15}, and are further supported by neuroimaging evidence showing atypical activation and connectivity in regions involved in facial expression processing, such as the mirror neuron system and amygdala^{9,16}. Together, these findings indicate that facial expression patterns offer a meaningful, noninvasive marker of ASD-related socio-emotional differences, underscoring their potential for aiding ASD identification and the need for objective, systematic evaluation of facial expression dynamics.

Notably, the subtle and context-dependent facial expressions often observed in ASD are difficult to capture through routine clinical assessment. Current diagnostic practice relies on expert ratings and standardized behavioral scales^{17,18}, which are subjective and based on relatively brief observation periods. These approaches require clinicians to summarize

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rapidly fluctuating social signals into broad categories, making it challenging to detect transient, low-intensity, or ambiguous facial behaviors. As a result, fine-grained expression differences may be underreported, and assessments can vary across evaluators. This gap between traditional observation and the level of precision needed to characterize facial expression dynamics underscores the importance of developing digital, objective methods for more consistent and detailed facial behavior analysis.

In response to the limitations of clinician-based observation, recent research has increasingly adopted digital, computer-vision-driven methods to characterize facial behavior in ASD. These approaches have primarily focused on discrete facial responses that are intentionally elicited under scripted, task-driven conditions. One prominent line of work involves expression-production or imitation tasks in which participants are instructed to pose specific emotions. Manfredonia et al. examined six discrete posed emotions in individuals aged 6–63 years and identified altered facial muscle activation in ASD¹⁹. Guha et al. analyzed children aged 9–14 years as they imitated preset target expressions and reported reduced eye-region movement complexity in the ASD group¹¹. A second line uses standardized emotional or social stimuli to elicit discrete, stimulus-driven facial responses. Samad et al. studied children and adolescents aged 7–20 years using 3D stimulus-based facial-expression tasks and found atypical smile patterns and increased gaze aversion in ASD²⁰. Sapiro et al. developed mobile-based assessments for toddlers aged 17–36 months, using brief scripted videos, name-calling prompts, and simple games to evoke discrete facial and behavioral responses, and showed that smartphone-captured features could support scalable ASD screening^{18,21–23}. A third line relies on structured question-response or diagnostic-style procedures to elicit discrete, guided facial responses. Drimalla et al. analyzed adults in a scripted simulated-dialogue task and found reduced smiling and diminished facial mimicry in ASD^{24,25}. Zhang et al. analyzed adolescents and adults aged 16–37 years during ADOS-derived interview segments and showed that spatio-temporal facial features extracted from standardized responses supported accurate classification²⁶. Sariyanidi et al. used short conversational segments from the CASS social-skills task and found that facial-movement time series enabled AI-based diagnostic predictions comparable to human raters²⁷. Zhao et al. studied children aged 6–13 years in scripted social tasks and found that eye-tracking and head-movement patterns elicited under guided testing conditions provided diagnostic value for ASD^{28,29}. In addition, several studies have extended these analytic strategies to brief, predefined interaction sequences. Georgescu et al. extracted interpersonal synchrony features from task-defined dyadic interactions³⁰, and Koehler et al. derived reciprocity measures from instructed conversational exchanges³¹. Although these studies broaden the types of elicited behavior under examination, they remain anchored in predefined or otherwise constrained interaction structures and continue to rely on discrete, analyzable behavioral segments that limit ecological validity.

Scripted paradigms have shown diagnostic value, but their reliance on contrived settings introduces several limitations. Because each trial is tied to predefined emotional prompts or highly structured tasks, these methods elicit only brief, predetermined responses and discrete expression events. Such designs oversimplify the continuous and context-dependent nature of social interaction, which involves blended affective states and subtle emotional transitions that scripted stimuli cannot capture. The lack of authentic social context further limits interpretability, as facial behavior may reflect task compliance rather than naturalistic socio-emotional processing, which is central to ASD-related social difficulties and their identification. Scripted tasks are also monotonous and cognitively demanding, often requiring specialized guidance, which reduces their suitability for young children and limits their potential for scalable screening^{26,32,33}. In terms of population coverage, most studies focus on adults, adolescents, or toddlers, leaving children aged three to ten years, a critical developmental period for early identification^{34–36}, noticeably underrepresented. Methodologically, existing work emphasizes static, predefined expressions and short temporal windows, frequently overlooking sustained emotional fluctuations, temporal dynamics, and coordination across facial muscles. These features are

essential for characterizing the atypical or poorly modulated expressions commonly reported in ASD¹¹. Together, these limitations highlight a gap between task-elicited measurement and the objective characterization of facial behavior as it unfolds in real social contexts.

To address these limitations, this study examined facial expression dynamics in children with ASD during naturalistic interactions. Naturalistic settings allow children to engage in spontaneous, self-initiated behavior without predefined emotional cues or task constraints. As a result, facial expressions emerge in familiar, ecologically valid contexts where social communication difficulties typically manifest. Such unscripted interactions also reveal a wider repertoire of affective behavior, including subtle, mixed, and rapidly shifting emotional states that structured tasks rarely capture. These characteristics provide a richer basis for identifying expressive patterns that are clinically meaningful and more reflective of everyday social functioning.

In this study, we analyzed 184 naturalistic interaction sessions from children with ASD and typically developing (TD) peers aged three to ten years. From these videos, we quantified three features of spontaneous facial expression: emotion variation, reflecting the temporal stability of emotional states; expression intensity, capturing the magnitude of facial muscle activation; and facial coordination, representing the synchrony across facial muscles. These features offered holistic and temporally sensitive representations across coarse and fine scales. They enable detailed quantification of atypical expressive patterns and support downstream ASD classification and symptom severity assessment. By grounding measurement in unconstrained, developmentally appropriate, and socially meaningful interactions, this framework addresses key limitations of scripted paradigms and provides a more ecologically valid characterization of ASD-related facial behavior.

Results

Study details and participant characteristics

The workflow for multidimensional facial expression dynamics analysis in naturalistic interactions for ASD assessment is illustrated in Fig. 1. Children participated in unscripted interactions to elicit spontaneous facial expressions, and clinical assessment scores were collected as reference measures using the Autism Behavior Checklist (ABC) and the Clancy Autism Behavior Scale (CABS). Three core dimensions of facial behavior were extracted, including emotion variation, expression intensity, and facial coordination, and these were used in subsequent statistical and machine learning analyses. A total of 186 children participated in the study, with two excluded for wearing masks, resulting in 99 children with ASD and 85 TD children. Demographic and experimental characteristics are summarized in Table 1. Independent-samples t-tests indicated no group differences in age or session duration ($p > 0.10$), supporting comparable baseline conditions. Consistent with the higher prevalence of ASD in males³⁷, the ASD group included a larger proportion of male participants. Follow-up MANOVA analyses further examined this imbalance and confirmed that it did not influence the observed group differences (see Supplementary Table 5). Clinical assessments indicated that most children in the ASD group were classified by clinicians as having mild-to-moderate symptom severity. Mann-Whitney U-tests showed significant differences in ABC and CABS scores ($p < 0.01$), reflecting clear distinctions in symptom severity between the ASD and TD groups.

Emotion variation Differences Between Groups

Group comparisons of emotion variation during naturalistic interactions were conducted based on global explained variance, occurrence frequency, duration, and emotion transition, as shown in Fig. 2. These metrics provide a coarse-grained characterization of emotional dynamics, with global explained variance, frequency, and duration reflecting holistic descriptive features, while the transition matrix captures dynamic patterns. No significant differences were observed between ASD and TD groups in neutral, happy, surprise, and sad emotions across these metrics ($p > 0.05$). However, the ASD group exhibited significantly higher global explained variance,

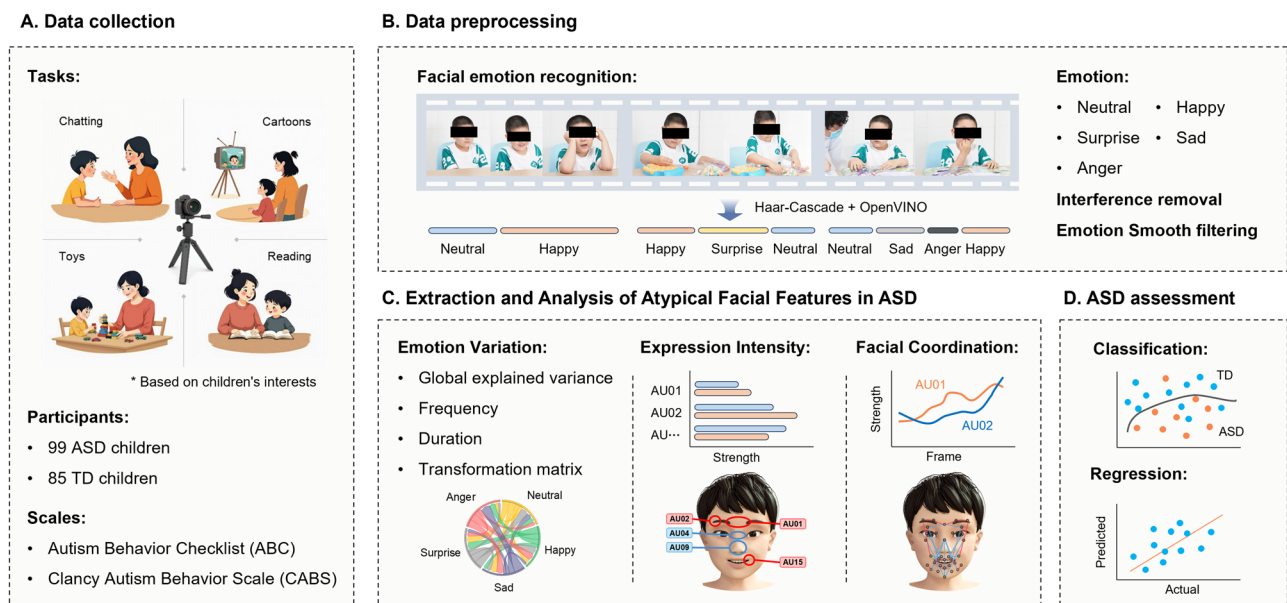


Fig. 1 | Workflow of multidimensional facial-expression dynamics analysis in naturalistic interactions for ASD assessment. **A** A total of 186 children engaged in naturalistic interactions while being video recorded; two were excluded due to mask-wearing, leaving 184 usable sessions from 99 ASD and 85 TD children, with ABC and CABS scores collected. **B** Facial regions were detected and frame-wise emotions

(neutral, happy, surprise, sad, anger) were recognized, followed by interference removal and temporal smoothing to obtain stable emotion trajectories. **C** Three spontaneous expression features, including emotion variation, expression intensity, and facial coordination, were extracted and analyzed for group differences. **D** The fused features were used for ASD classification and symptom severity prediction.

Table 1 | Demographic and experimental characteristics

	ASD group	TD group	Stat. test	p value
Sample size	99	85	-	-
Age (month)	75.93 ± 17.99	78.41 ± 18.88	-0.825	0.410
Gender (male/female)	83/16	47/38	16.621	< 0.001
ABC scale	38.43 ± 23.48	1.72 ± 3.49	-11.584	< 0.001
CABS scale	10.75 ± 5.39	1.13 ± 1.62	-10.949	< 0.001
Session duration (minute)	24.28 ± 4.45	23.18 ± 4.02	-1.748	0.101

Continuous variables are presented as mean ± standard deviation (SD), and categorical variables are presented as counts.

occurrence frequency, and duration in the anger state compared to the TD group ($p < 0.01$), suggesting an increased prominence of anger-related expressions in ASD. Analysis of the emotion transition matrix revealed distinct alterations in transition probabilities within the ASD group. Specifically, the probability of transitioning from sad to neutral was significantly reduced (-6.7% , $p = 0.03$), as was the transition from happy to surprise (-7.8% , $p = 0.02$). In contrast, transitions from anger to sad were significantly increased ($+12.3\%$, $p = 0.03$). Additionally, except for surprise, transitions from all other emotions to anger were significantly more frequent in the ASD group ($p < 0.05$), indicating a heightened tendency for anger-related emotional shifts. Detailed numerical values for both the ASD and TD transition matrices are available in Supplementary Table 1 and Table 2.

Expression intensity differences between groups

Fig. 3 shows the group differences in expression intensity across emotional states. As a fine-grained feature, expression intensity reflects the average activation of individual facial muscles, enabling detailed analysis beyond emotional categories. The dominant AU combinations under different emotional states were largely consistent between the ASD and TD groups. For instance, in the happy condition, AU 06, 07, and 25 were prominently activated, while in the sad condition, AU 07 and 17 were the key contributors to facial expression. However, across all emotional states, the ASD group showed significant AU intensity differences compared to the TD

group, with these differences primarily characterized by higher AU intensity in the ASD group ($p < 0.05$). This suggests exaggerated or atypical emotional expressions in individuals with ASD. Notably, the significant increases in AU intensity observed in the ASD group across all emotional states were mainly occurred in non-core AUs, particularly those with lower baseline intensity. This pattern suggests that individuals with ASD tend to rely more on these non-core AUs, and the over-activation of these less prominent components may contribute to the ambiguity and atypicality of their facial expressions.

Facial coordination differences between groups

Fig. 4 shows the differences in facial AU coordination between the ASD and TD groups across five emotional states. Facial coordination reflects the degree to which pairs of facial muscles vary together over time. The ASD group exhibited altered coordination patterns, with both higher and lower coordination values observed across different AU pairs ($p < 0.05$). Although changes appeared in both directions, a larger number of AU pairs showed increased coordination than decreased coordination, indicating an overall tendency toward stronger coupling among certain facial muscles in ASD. These differences were most evident in neutral and sad expressions, where lower arousal levels and smaller movement amplitudes may make coordination changes easier to detect. Moreover, the ASD group showed a pronounced increase in vertical connectivity rather than localized facial regions, indicating atypical coordination between upper and lower facial areas. This pattern may reflect subtle stereotyped behaviors and is consistent with clinical descriptions of facial expressions in ASD as atypical or poorly modulated.

Classification performance of ASD

To evaluate the effectiveness of different feature representations in distinguishing ASD from TD individuals, classification experiments were conducted using emotion variation, expression intensity, and facial coordination features, both individually and in combination. The t-SNE visualizations (Fig. 5) illustrate feature distributions for the ASD and TD groups across different fusion strategies. When using only emotion variation features, the data points for both groups largely overlap, indicating limited discriminatory power. However, the inclusion of expression intensity and

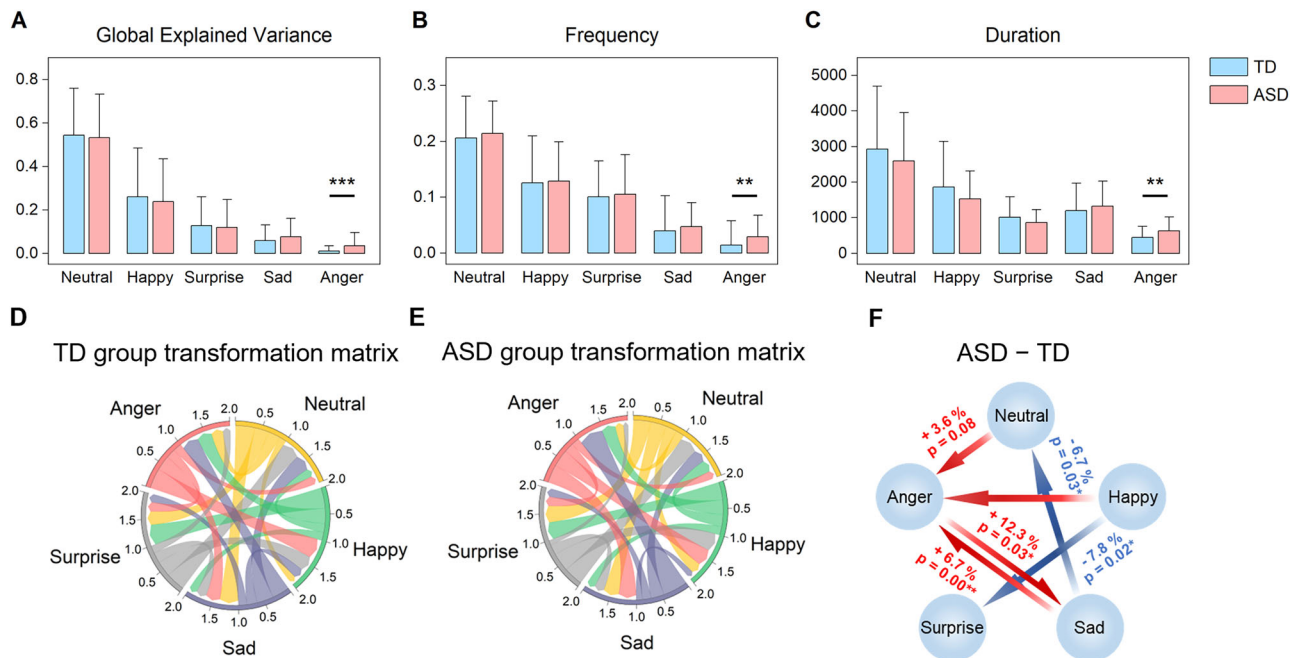


Fig. 2 | Group differences in emotion variation during naturalistic interactions. **A** Global explained variance of each emotion for ASD (red) and TD (blue) groups, reflecting the overall distribution and stability of emotional states across time. **B** Occurrence frequency and **(C)** average duration of emotional episodes for each emotion in ASD and TD children. Error bars in **(A–C)** indicate standard deviations. **D, E** Emotion transition matrices for the TD group **(D)** and ASD group **(E)**. For each emotion, the 0–1 segment of the arc represents the proportion of transitions leaving that emotional state, and the 1–2 segment represents the proportion of transitions

entering that state. Chord widths indicate transition probabilities between emotion pairs. **F** Between-group differences in transition probabilities, expressed as ASD minus TD (ASD–TD). Red arrows show transitions that occur more frequently in the ASD group, whereas blue arrows indicate transitions that occur less frequently than in TD children. Labels report percentage changes and corresponding *p*-values. Asterisks (*, **, ***) denote significant group differences ($p < 0.05$, $p < 0.01$, $p < 0.001$).

Table 2 | ASD classification performance based on atypical facial expression patterns

Feature	Sensitivity	Specificity	PPV	NPV	Accuracy	F1-score	AUC
EV	0.758	0.682	0.735	0.707	0.723	0.746	0.790
Fusion EV-EI	0.828	0.882	0.891	0.815	0.853	0.859	0.941
Fusion EV-FC	0.909	0.812	0.849	0.885	0.864	0.878	0.925
Fusion EI-FC	0.949	0.776	0.832	0.930	0.870	0.887	0.936
Fusion EV-EI-FC	0.960	0.882	0.905	0.949	0.924	0.931	0.977

EV Emotion variation, EI Expression Intensity, FC Facial Coordination. Bold values indicate the best performance.

facial coordination enhanced group separation. The clearest separation was observed with full feature fusion, where ASD and TD distributions showed minimal overlap, indicating improved group distinction. This trend is also reflected in Table 2. Using only emotion variation features yielded the classification accuracy 72.3% and AUC 0.790. Feature fusion significantly improved performance, with two-feature combinations outperforming single-feature models. The highest performance was achieved with full feature fusion (emotion variation, expression intensity, and facial coordination), resulting in accuracy 92.4%, F1-score 0.931, and AUC 0.977. These results were obtained using an support vector machine (SVM) classifier tuned to a regularization strength of 0.1, a kernel width corresponding to the scale setting, and a second-degree polynomial kernel with a coef0 value of 1. These findings demonstrate that combining emotion variation, expression intensity, and facial coordination provide the most robust classification of ASD, highlighting the importance of incorporating multiple facial expression features for improved accuracy and reliability in ASD identification.

Regression prediction of ASD severity

The regression prediction results for ASD severity, based on facial expression features, exhibited a similar trend for both the ABC and CABS scales (Table 3). Using only emotion variation yielded relatively higher RMSE and

MAE values, with an R^2 of 0.157 for the ABC scale and 0.145 for the CABS scale. However, incorporating additional features improved performance across all metrics. The best results for both scales were achieved when combining emotion variation, expression intensity and facial coordination features, with the ABC scale showing an RMSE of 18.98, MAE of 13.94, and an R^2 of 0.432, and the CABS scale showing an RMSE of 4.88, MAE of 3.84, and an R^2 of 0.400. These findings highlight that fusing coarse-grained emotion variation with fine-grained expression intensity and facial coordination substantially improves the accuracy of ASD severity prediction for both scales. Scatter plots of the true versus predicted scores for both the ABC and CABS scales (Fig. 6) further illustrate the enhanced prediction performance, showing an evident correlation between actual and predicted values following the fusion of fine-grained facial dynamic features.

Robustness checks

To ensure that these findings were not driven by data-quality issues or processing artifacts, we conducted the post-hoc robustness checks (see Methods). Usable-frame rates after preprocessing quality control were comparable between groups, and head-movement deviations showed no systematic differences (supplementary Fig. 1 and Table 4), indicating that neither frame visibility nor motion-related noise differentially biased feature

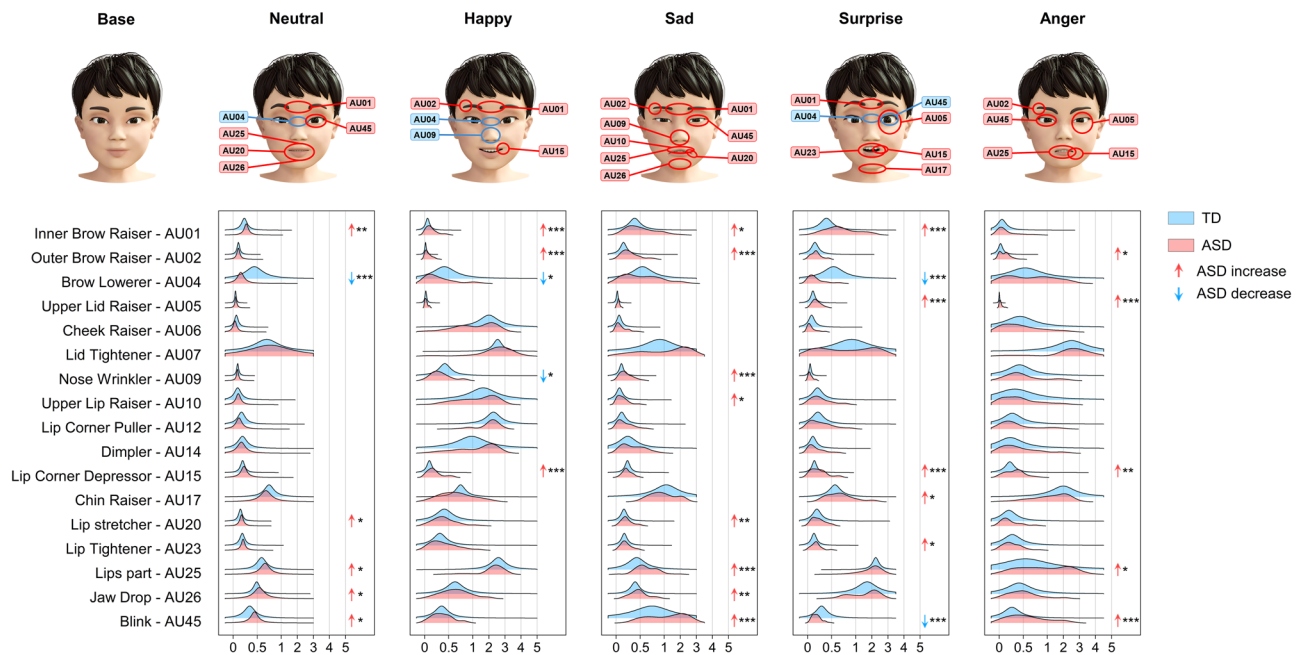


Fig. 3 | Group differences in facial expression intensity across emotional states (Neutral, Happy, Sad, Surprise, and Anger). For each emotion category, the lower panels display the intensity distributions of each Action Unit (AU) for the TD (blue) and ASD (red) groups. The vertical axis lists each AU together with its corresponding facial muscle behavior, and the density curves illustrate the magnitude and variability of AU activation. Red and blue arrows indicate statistically significant

increases or decreases in AU intensity in the ASD group relative to the TD group, respectively, with asterisks denoting significance levels. The top panel provides a visual reference for the anatomical locations of AUs that show significant group differences. Red circles mark AUs with significantly higher intensity in ASD, whereas blue circles mark AUs with significantly lower intensity compared with TD children.

Fig. 4 | Differences in facial coordination between the ASD and TD groups across emotional states.

The lines represent significant AU coordination differences, with red indicating increased coordination in the ASD group and blue indicating decreased coordination compared to the TD group.

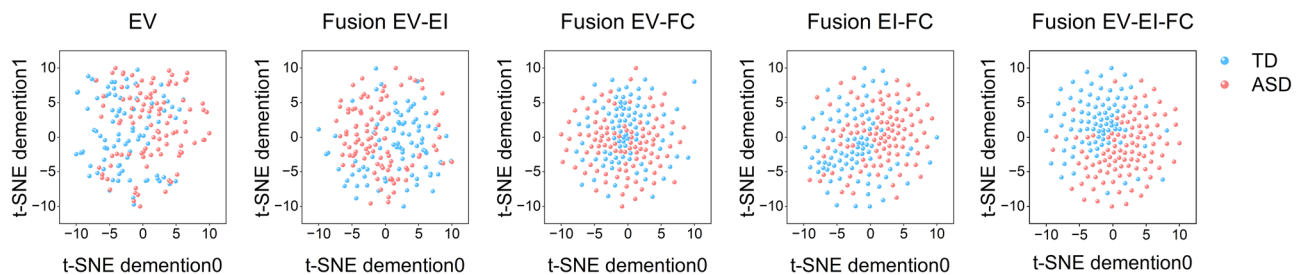
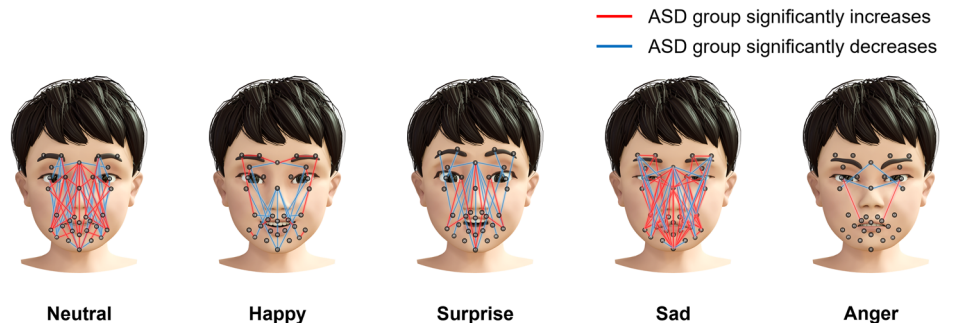


Fig. 5 | The t-SNE visualizations of feature distributions for ASD and TD groups across different fusion strategies. EV Emotion variation, EI Expression Intensity, FC Facial Coordination.

extraction. We also examined potential gender influences and found that gender did not account for additional variance in any facial-expression feature domain (supplementary Table 5). Together, these checks support the reliability of the extracted features and provide a robust foundation for the group-level and predictive findings.

Discussion

We analyzed facial expression behaviors in children with ASD during unscripted naturalistic interactions, focusing on emotion variation, expression intensity, and facial coordination. First, this study is the first to quantitatively characterize atypical expression patterns in naturalistic

settings using a multidimensional feature set that integrates holistic and processual components at coarse- and fine-grained levels. Second, children with ASD exhibited distinctive profiles, including increased prominence of anger, altered emotion transition probabilities, heightened intensity in non-core action units, and atypical facial coordination, reflecting atypical expressive mechanisms. Third, these features distinguished ASD from TD children and predicted symptom severity, highlighting their potential clinical value for early identification and assessment.

Traditional ASD-related facial analysis studies have relied on discrete or task-driven measures that require subjects to respond to predefined cues, as summarized in Table 4. These paradigms often fail to elicit spontaneous, context-dependent emotional expressions, particularly in children who may suppress affect in artificial settings. In contrast, our approach captures facial dynamics during unscripted, naturalistic interactions, revealing ecologically valid spontaneous cues that differ from posed or elicited emotions in their temporal and intensity patterns^{19,38,39}. Although our approach does not rely on task-dependent assessment, a supplementary comparison of manually annotated activity segments showed that classification performance was highly consistent across different task contexts, indicating that the behavioral signals extracted from naturalistic interactions were robust and not dependent on specific tasks (see Supplementary Table 3). Building on this, and despite cohort differences that limit direct comparison with prior studies, the relatively high recognition performance observed in our setting further indicates that authentic interactions may convey stronger diagnostic information than scripted tasks. This perspective is also supported across other modalities; for example, studies on speech-based ASD recognition have shown that spontaneous vocal expressions are critical for identifying atypical speech patterns^{26,32,40}. Moreover, leveraging naturalistic interactions

reduces the cognitive demands of scripted tasks, making the assessment setting more accessible and developmentally appropriate for young children. Finally, this non-invasive, passive, and scalable approach is suitable for real-world environments, such as homes and schools, providing a practical strategy for early ASD screening and continuous behavioral monitoring while addressing limitations of existing diagnostic practices.

Our findings demonstrate that the dynamics of emotional state changes provide a sensitive marker of the social interaction and emotion regulation difficulties observed in children with ASD. Compared with TD peers, ASD children exhibited greater challenges in modulating facial expressions in response to social cues, reflecting heightened emotional instability and irritability. This pattern aligns with experiential impressions that ASD individuals are prone to sudden emotional outbursts¹³. The increased variability and intensity of their facial expressions further indicate difficulties in emotional self-regulation. Across emotional contexts, ASD children generally showed elevated AU activation, suggesting exaggerated or atypical expressions that may stem from atypical neuromuscular control or compensatory mechanisms during emotion production^{41–43}. Importantly, they relied more heavily on non-core AUs, whose heightened activation reduced the clarity and precision of expressions. This reliance aligns with behavioral reports noting that facial expressions in ASD can be difficult to characterize or interpret⁴⁴. Limited fine motor control may further contribute to this ambiguity, highlighting the role of nonverbal motor mechanisms in social communication difficulties^{45,46}. In addition, atypical coordination patterns in ASD children were especially pronounced in vertically connected facial regions rather than isolated local areas. This reduced synchrony between upper and lower facial regions may lead to expressions that appear less natural or show atypical coordination and may relate to subtle repetitive behaviors or rigid motor tendencies often observed in ASD. Together, these findings extend prior evidence of atypical facial expressivity in ASD^{11,47,48} by demonstrating how disruptions in expression dynamics contribute to challenges in social interaction.

Emotional instability, including heightened irritability and difficulty regulating emotions, is a well-recognized trait of ASD^{11–13} and is commonly assessed in clinical evaluations. In our study, emotion variation (EV) features aligned with this trait, but alone yielded limited classification performance (accuracy 72.3%, AUC 0.790), indicating that broad emotional fluctuations are insufficient to distinguish ASD from TD children. To capture more nuanced differences, we incorporated expression intensity (EI) and facial coordination (FC). The fusion of EV and EI (Fusion EV-EI) substantially improved classification (accuracy 85.3%, AUC 0.941), reflecting that ASD children exhibit distinct facial muscle activation patterns that extend beyond general emotional instability. Adding FC (Fusion EV-EI-FC) further enhanced performance (accuracy 92.4%, AUC 0.977), capturing atypical intensity control and coordination deficits. These results suggest that ASD-related facial expressions are not only unstable but also poorly coordinated, producing disjointed or less fluid expressions. Such deficits are often overlooked in traditional observational assessments, which

Table 3 | Regression prediction performance for ASD severity based on atypical facial expression patterns

Scales	Feature	RMSE	MAE	R ²
ABC	EV	23.13	17.76	0.157
	Fusion EV-EI	20.29	14.79	0.351
	Fusion EV-FC	20.22	14.74	0.355
	Fusion EI-FC	19.57	14.50	0.396
	Fusion EV-EI-FC	18.98	13.94	0.432
CABS	EV	5.82	4.60	0.145
	Fusion EV-EI	5.19	4.02	0.320
	Fusion EV-FC	5.05	4.01	0.358
	Fusion EI-FC	4.99	3.94	0.371
	Fusion EV-EI-FC	4.88	3.84	0.400

EV Emotion variation, EI Expression Intensity, FC Facial Coordination. Bold values indicate the best performance.

Fig. 6 | Prediction performance of ASD severity scores based on multimodal facial features. Scatter plots illustrate the relationship between actual and predicted ASD severity scores for the ABC and CABS scales, based on the fusion of emotion variation, expression intensity, and facial coordination. Each point represents one participant.

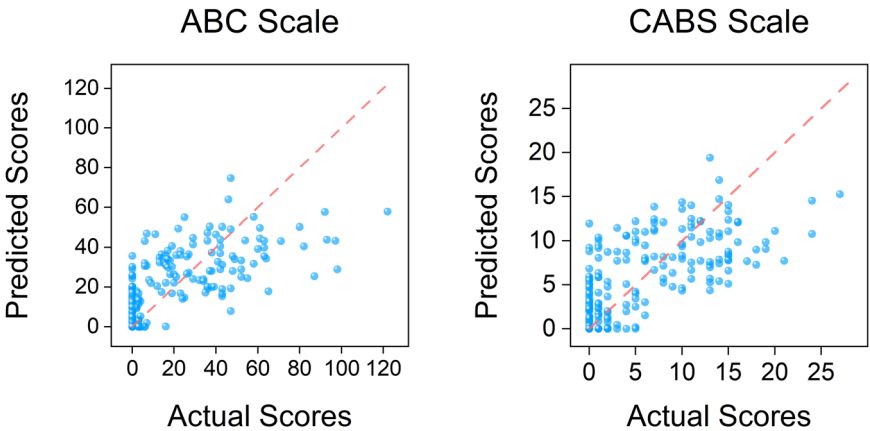


Table 4 | Comparison with Representative ASD Studies Using Facial Analysis

Reference	Cohort Size	Assessment Protocol	Metrics
Drimalla et al. ²⁴	80 adults (37 ASD / 43 TD)	scripted simulated-dialog	accuracy = 73%, AUC = 0.74
Koehler et al. ³¹	88 adults (28 ASD / 60 TD)	two structured topic discussions	accuracy = 79.5%, AUC = 0.82
Cheng et al. ¹⁸	95 children (71 ASD / 24 TD)	structured, task-based interaction	accuracy = 88.42%
Hashemi et al. ²²	33 toddlers (15 ASD / 18 TD)	predefined movie stimuli	precision = 0.89, F1-score= 0.89
Zhang et al. ²⁶	33 adults (Autism / Autism Spectrum / Non-Spectrum)	ADOS professional assessment video	accuracy = 91.72%
Ours	184 children, 3–10 years (99 ASD / 85 TD)	unscripted naturalistic interaction	accuracy = 92.4%, AUC = 0.977

focus on broad emotional states rather than the underlying dynamics of facial movements^{18,23,49}. Our findings highlight that analyzing facial expression patterns beyond coarse-grained emotional variation provides more precise and objective insights into ASD-related atypicality.

Beyond classification, the Fusion EV-EI-FC model achieved meaningful performance in predicting ASD symptom severity, with mean absolute errors of 13.94 points for the ABC scale and 3.84 points for the CABS scale, indicating that facial-expression features contain severity-related information. However, the model showed reduced accuracy for children with more severe symptoms, and the R^2 values (40%–43%) suggest that facial expressions provide only a moderate level of explanatory power. Several factors may account for these outcomes. First, the severity distribution in our dataset resembled typical clinical populations, with most children classified as mild to moderate and very few high-severity cases available for model learning, limiting the model’s ability to generalize to the most impaired group. Second, caregiver-reported scale scores are often influenced by limited understanding of item descriptions or under-reporting, which is commonly observed in real-world assessments and can reduce the consistency with which scale values reflect true symptom levels^{50,51}. Third, facial expressions convey only about 55 percent of a person’s social information^{7,8}, and ASD symptoms show substantial heterogeneity, which may restrict the extent to which facial-expression features alone can capture continuous variations in symptom severity. Together, these factors suggest that while the proposed method supports effective classification, their capacity to accurately predict symptom severity remains limited and warrants cautious interpretation.

Considering these points, several limitations of the present study should be noted. First, the generalizability of our method would benefit from validation in larger and more diverse samples to confirm its robustness across different populations and settings. Second, the current framework uses only facial-expression features, which reflect just one aspect of social behavior. Given the heterogeneity of ASD, future work incorporating additional behavioral information may help provide a more complete description of symptom characteristics and further improve estimation performance.

In summary, this study presents an ecologically valid approach for ASD assessment through the analysis of spontaneous facial expressions during naturalistic interactions. Integrating emotion variation, expression intensity, and facial coordination, this method captures both holistic and processual features at coarse- and fine-grained levels, revealing atypical patterns such as unstable emotions, heightened intensity, and altered coordination that differentiate ASD from TD children. Importantly, it uncovers subtle facial dynamics inaccessible to traditional approaches, providing a quantitative explanation for the hard-to-describe atypical expressions often observed in ASD. The approach achieved meaningful

classification and symptom-severity prediction, providing a potential basis for objective early screening without structured tasks or professional supervision, and supporting feasibility in real-world applications.

Methods

Overview

Figure 1 illustrates the process of quantifying facial expression dynamics in children with ASD during naturalistic, unscripted interactions. During data collection, children engaged in free conversational activities to elicit spontaneous expressions, and standardized clinical assessment scores were obtained as reference measures. The preprocessing stage employed facial emotion recognition and refinement procedures to generate temporally consistent emotion sequences. Feature extraction focused on three dimensions of facial behavior: emotion variation, expression intensity, and facial coordination, capturing both holistic and processual aspects of expression. Statistical analyses were then performed to compare the ASD and TD groups. Finally, machine learning models were applied for ASD assessment, including diagnostic classification and symptom severity estimation. The corresponding code is publicly available on GitHub: <https://github.com/FromNature/AutismFacialDetection>.

Data collection

This study was part of a cross-sectional clinical investigation on behavioral characteristics of ASD. Participants were recruited from multiple sources to ensure representativeness. Children with ASD were diagnosed at Children’s Hospital, Tianjin University, and TD children were recruited from hospital check-ups, local schools, and nearby communities to ensure an unbiased sample. Inclusion criteria were: (a) ASD diagnoses confirmed by experienced clinicians based on DSM-5 and ICD-10 criteria^{52,53}, while TD children had no history of ASD or other psychiatric disorders; (b) age between 3 and 10 years; (c) absence of recent conditions such as colds, diarrhea, or pain that could affect behavior; and (d) normal vision and hearing in both groups. Ethical approval for this study was obtained from the Research Ethics Committee of Tianjin Children’s Hospital (Approval No. 2023-IITKY-002) prior to the commencement of the experiment, in accordance with the Declaration of Helsinki. Written informed consent was voluntarily provided by the parents or legal guardians of all participating children, and verbal assent was also obtained from the children prior to their participation.

Data collection took place in a quiet private room with assistance from a parent or a familiar caregiver. To foster naturalistic engagement, the environment was arranged to resemble typical home or school settings, with multiple thematic materials available throughout the session, including toys, picture books, and cartoons on a television. The facilitator interacted with each child in an unscripted manner, following the child’s interests and allowing spontaneous activity choices rather than imposing any fixed sequence. Sessions were planned for about 30 min, although the actual duration varied with the interaction context. This setup was intended to encourage naturalistic interaction and emotional expression. A Sony ZV-E10 camera positioned at an appropriate distance provided a stable frontal view and continuously recorded the child’s upper body at 1920 × 1080 resolution and 50 fps, while the facilitator used non-directive interaction materials or subtle cues only when necessary to help the child maintain an appropriate viewing angle without restricting spontaneous behavior. Clinical severity was assessed using two validated parent-report instruments: the Autism Behavior Checklist (ABC; 57 items, total score 158) and the Clancy Autism Behavior Scale (CABS; 14 items, total score 28), where higher scores indicate greater symptom severity.

Data preprocessing

Facial detection and alignment were performed using the confidence-based Haar-Cascade algorithm in OpenCV to extract facial regions^{54,55}. Emotion recognition was then conducted using the emotions-recognition-retail model, a pretrained convolutional neural network from the Intel Open Model Zoo that provides high-accuracy classification across five discrete emotions (neutral, happy, surprise, sad, and anger)⁵⁶. The model was

executed through the OpenVINO backend to enable efficient processing of long, high-frame-rate videos and to support deployment on mobile or resource-limited devices⁵⁷. Frames with insufficient facial information were labeled as “undefined” and excluded. Importantly, the subsequent analysis did not rely on this specific model because the preprocessing pipeline only requires frame-wise emotion label trajectories from any reliable classifier. To preserve genuine emotional transitions while suppressing frame-level noise, we applied a 200 ms sliding-window filter that removed predictions shorter than this duration. This parameter was selected based on micro-expression research indicating that meaningful emotional changes typically last at least 200 ms^{58,59}. The smoothed emotion labels were then converted into temporal sequences describing the onset, offset, and duration of each emotional state, capturing the dynamic evolution of facial-expression patterns.

Facial feature extraction

Children with ASD are often described clinically as irritable and prone to agitation, underscoring the importance of emotion-related characteristics for understanding their behavioral patterns. To quantify these characteristics, we analyzed emotion variation features during unscripted interactions, providing a coarse-grained description of overall emotional dynamics, including both holistic emotional states and their transitions. Beyond these general fluctuations, children with ASD frequently show reduced naturalness or atypically coordinated facial expressions, which are difficult to capture objectively. To address this, we extracted expression intensity features that index the activation of specific facial muscles, as well as facial coordination features that characterize dynamic relationships among muscle movements and offer a finer-grained view of facial expression patterns. Because facial movements are closely linked to emotional states, emotion-state alignment was incorporated into both feature types to enhance precision and interpretability. The following paragraphs describe the extraction procedures for emotion variation, expression intensity, and facial coordination.

The emotion variation features included multiple complementary properties of emotional expression, covering global explained variance (GEV), frequency, duration, and transition probabilities. To characterize emotional fluctuations in ASD, we adopted an approach inspired by microstate analysis, a technique commonly used to study neural dynamics and psychological states^{60–62}. This approach enables the identification of distinct, transient states over time, making it particularly well-suited for capturing the dynamics of emotional expression. In this study, five discrete emotional states were considered: neutral, happy, surprise, sad, and anger.

The GEV of each emotion across the entire interaction was quantified to reflect the overall distribution of emotional states. This can be expressed as:

$$GEV_{emotion} = \frac{n_{emotion}}{N_{total}}$$

where $n_{emotion}$ represents the number of frames classified as a specific emotional state, and N_{total} denotes the total number of frames in the entire interaction.

Additionally, the occurrence frequency $f_{emotion}$ and average duration $d_{emotion}$ of each emotion were calculated. The $f_{emotion}$ represents the emotion occurred per second, computed as:

$$f_{emotion} = \frac{C_{emotion}}{N_{total}/f_s}$$

where $C_{emotion}$ represents the total number of occurrences of the specific emotional state during the interaction, and f_s is the sampling rate of the video (frames per second).

The $d_{emotion}$ was calculated to reflect the typical length of time a specific emotion was maintained before transitioning to another state. This was

defined as:

$$d_{emotion} = \frac{n_{emotion}}{C_{emotion}}$$

To explore the transitions between emotions, an emotion transition matrix P , which encodes the probability of shifting from one emotional state to another. The element $P(i,j)$ represents the probability of transitioning from emotional state i to state j , and is computed as:

$$P(i,j) = \frac{C(i,j)}{\sum_k C(i,k)}$$

where $C(i,j)$ is the count of transitions from state i to state j , and the denominator is the total number of transitions from state i to any other state. Notably, transitions from a state i to itself (or to the same emotion) were not included in the calculation of transition probabilities. This is because such self-transitions tend to dominate and are highly frequent, making them less informative. Additionally, the information provided by these self-transitions overlaps significantly with the average duration feature, which captures the persistence of each emotional state.

Grounded in the principles of microstate analysis, the emotion variation features provide a structured, multi-perspective summary of emotional behavior. They integrate coarse-grained, holistic indicators of emotional distributions and durations with dynamic transition patterns, facilitating the interpretation of spontaneous emotional fluctuations in children with ASD.

Expression intensity features were used in this study as fine-grained measures of subtle facial muscle activation. Unlike coarse-grained emotional states that describe overall affective tendencies, these features capture micro-level variations in facial muscle movements^{63,64}, which are often atypical in children with ASD and may manifest as reduced or exaggerated expressions. Quantifying such variations provides an objective basis for assessing differences between ASD and TD groups.

The Facial Action Coding System (FACS)^{65,66}, based on anatomical principles, was employed for feature extraction. FACS represents facial expressions as combinations of discrete Action Units (AUs), each corresponding to the activation of specific muscles or muscle groups. By quantifying AU intensities, FACS provides a systematic approach to decomposing facial movements, allowing a more detailed analysis of expression patterns beyond broad emotional categories.

AU intensity features were extracted using OpenFace for automated facial analysis. The pipeline included facial detection, landmark alignment, and AU intensity estimation. Seventeen AUs were quantified, each representing the activation strength of a specific facial muscle on a continuous scale from 0 to 5. Because emotional expressions are closely linked to AU activation patterns, the intensities were grouped according to corresponding emotional states. For each state, the average intensity of each AU was calculated, providing a holistic, state-level representation of facial muscle activation. This approach enables a structured and interpretable quantification of atypical facial expressions in children with ASD.

Facial coordination features were defined in this study as the temporal synchronization among AU intensity trajectories within each emotional state. Facial expressions arise from the interplay of multiple facial muscles, and certain AUs tend to co-activate in characteristic ways during specific emotions; for example, AU 02 (outer brow raiser) and AU 12 (lip corner puller) commonly appear together during happy expressions. By quantifying these temporal relationships, facial coordination features provide a process-level measure of how coherently facial muscles move together over time, offering a fine-grained characterization of expression patterns beyond emotion labels alone. This measure is particularly relevant for identifying atypical patterns in children with ASD, who often show reduced naturalness or non-fluent facial expressions.

For each participant, the intensity time series of AU *i* and AU *j* were extracted within a given emotional state. Let:

$$s_i = \{s_i(1), s_i(2), \dots, s_i(N)\}, s_j = \{s_j(1), s_j(2), \dots, s_j(N)\}$$

denote the AU intensity sequences of AU *i* and AU *j* over *N* frames within a specific emotional state. Facial coordination was computed using the Spearman rank correlation, which evaluates the monotonic relationship between two sequences and is robust to non-linear dependencies:

$$r(\text{AU } i, \text{AU } j) = \rho_{i,j} = \text{Spearman}(s_i, s_j) = 1 - \frac{6 \sum_{n=1}^N (R_i(n) - R_j(n))^2}{N(N^2 - 1)^2}$$

where $R_i(n)$ and $R_j(n)$ are the rank-transformed values of AU *i* and AU *j* at frame *n* and $\rho_{i,j}$ is the resulting correlation coefficient. Higher values indicate stronger coordination activation between the AU pair.

By quantifying AU synchronization patterns across different emotions, the facial coordination feature provides a fine-grained representation of facial expression dynamics and support detailed examination of atypical coordination patterns in ASD.

Statistical analysis

Statistical analyses were conducted to examine group differences across the three facial measures. For emotion variation features (global explained variance, frequency, duration, and transition probabilities), expression intensity features (AU intensities across emotion categories), and facial coordination features (AU pairwise coordination values), ASD-TD group comparisons were performed separately for each emotion. Distributional properties were first assessed for each feature. Normality was tested using the Shapiro-Wilk test and homogeneity of variance using Levene's test. When both assumptions were met, two-tailed independent-samples *t*-tests were conducted; otherwise, two-tailed Mann-Whitney *U* tests were applied. To control for Type I error from multiple comparisons, False Discovery Rate (FDR) correction was applied, with $p < 0.05$ being considered statistically significant. All analyses were conducted in SPSS (Version 27).

Robustness analyses were conducted to ensure that the observed group differences were not driven by data-quality issues, head movement artifacts, or gender-related imbalances. Frame-level data quality was quantified as the proportion of usable frames based on confidence thresholds from face detection, emotion recognition, and AU extraction, and group differences were tested using non-parametric methods. Head pose deviations in pitch, yaw, and roll were computed after baseline correction, and between-group differences were evaluated in the same manner. Gender-related effects were examined using a MANOVA with diagnostic group, gender, and their interaction as fixed factors; the three facial-feature domains were entered as dependent variables, Wilks' lambda was used as the multivariate statistic, and FDR correction was applied.

Additionally, to ensure that the sample size provided adequate sensitivity to detect meaningful group differences, statistical power was estimated using G*Power⁶⁷. The power analysis indicated a value of 0.92 for detecting medium-sized effects (effect size = 0.5, $\alpha = 0.05$, group sizes = 99 and 85), confirming that the study was sufficiently powered.

ASD assessment

To assess the usefulness of the extracted features for ASD identification, we performed both classification and regression analyses. Emotion variation features were used as the baseline because they provide a coarse-grained measure of overall expressive behavior, similar to the broad emotional descriptors used in clinical screening^{68,69}. Additional models combined emotion variation with expression intensity, facial coordination, or all three feature types to evaluate the contribution of fine-grained facial dynamics. Classification was conducted using a support vector machine (SVM). Hyperparameters were tuned through nested cross-validation, with an inner

grid search comparing linear, radial basis function, and polynomial kernels while exploring key parameters such as regularization strength, kernel width, and polynomial degree. Principal component analysis was applied within each training fold to avoid information leakage, and all features were normalized before training. Ridge regression was used to predict symptom-severity scores. Classification performance was evaluated using ten-fold cross-validation and reported using sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, F1-score, and AUC. Regression performance was assessed using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2).

Data availability

The video and clinical information cannot be made publicly available due to child privacy considerations but can be obtained from the corresponding authors upon reasonable request.

Code availability

To support reproducibility and future research, the source codes are openly available on GitHub: <https://github.com/FromNature/AutismFacialDetection>.

Received: 29 September 2025; Accepted: 13 January 2026;

Published online: 21 January 2026

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Acknowledgements

This research was supported by the Natural Science Foundation of Tianjin (Key Program, Grant No. 23JCZDJC01030; Grant No. 24ZXZSS00330) and the Autonomous Project of Haihe Laboratory of Brain-Computer Interaction and Human-Machine Integration (Grant No. 24HHNJSS00012; Grant No. 25HHNJSS00015).

Author contributions

M.D. designed the study and drafted the manuscript. P.S. performed data analysis and methodology. M.D., Z.L., and Y.X. contributed to algorithm

development. Z.L. and Y.X. were responsible for data collection and preprocessing. P.S. and X.L. assisted with data annotation and visualization. S.L. supervised the project and revised the manuscript. W.L. and D.M. provided overall guidance and final approval.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41746-026-02375-1>.

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